



**SRI SIVASUBRAMANIYA NADAR COLLEGE OF
ENGINEERING
DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

AN INTERNSHIP REPORT

AT



INDIA NIPPON ELECTRICALS LIMITED
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HOSUR-635109, TAMIL NADU

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The internship provided me with an excellent platform to connect the theoretical concepts learned during my academic studies with practical applications in the industry. The knowledge and experience gained during this period have strengthened my confidence and broadened my perspective on engineering and product development.

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ABSTRACT

This report is the collection of the knowledge and practical experience obtained from the internship performed at Indian Nippon Electricals Limited (INEL), Hosur, which is a well-known automotive electronic and electrical component manufacturer. This internship experience gave a deep insight in industrial, manufacturing, quality and developmental practices of automotive electronics industry. During this internship period, it was possible to acquire some understanding about various automotive electronic devices such as Capacitor Discharge Ignition(CDI), Transistor Controlled Ignition(TCI), Electronic Control Unit(ECU), Ignition Coil and Regulator Rectifier. The PCB designing and manufacturing techniques are explored; including surface mount technology(SMT), through hole technology(THT), soldering technique, quality control of the assembled PCBs and testing. Understanding of quality management systems and organizational procedures of industry along with practices to safeguard Electro Static Discharge(ESD) effects are gained. Furthermore, the working of motor controls system, selection of MOSFET for power applications and power electronic converters is studied. The technical learning aspect of the internship also includes the simulation of Zero Crossing Detector and Buck Converter with the help of LTspice so as to know their operational characteristics and their performances. It is sure that this internship was able to reduce the gap between academic knowledge and actual industrial operations and improve the problem solving skills and technical expertise for Electronics and Communication Engineering.

CHAPTER 1

INTRODUCTION

Industrial Training and Internship is very important in applying the knowledge that we learn from academics to real time applications. It enables the students to get practical experience about the working culture and develops the technical skills and also other qualities. The industrial training and internship is an essential part of the Bachelor of Engineering curriculum of Electronics and Communication Engineering in SSN College of Engineering. In my case, I have undergone an internship training at Indian Nippon Electricals Limited (INEL), Hosur. It is the leading automotive electronics and electrical component manufacturing firm. Products of INEL are distributed to all the major car manufacturers all over India. It is well-known in ignition system, engine management system, motor control systems and power electronic products for both the vehicles with 2 wheels, 3 wheels and electric vehicle. The working culture of INEL is a very good to learn. During the training period I have obtained knowledge of the processes followed for the design and fabrication of PCB and manufacturing of components, SMT, THT, quality management system, quality check, ESD precautions, motor control system, power electronic circuits such as the DC-DC Converter and MOSFET and other topics like the industrial norms followed and standards used in automotive electronic manufacturing and also the testing methods involved for testing the circuits. Also I have performed the simulation for some of the electronic circuits using the software LTspice and hence I could find the functioning of different circuits. I also performed the simulation for the zero crossing detector circuit. The above performed topics are helping in to find out the analysis and design concepts for the power electronics and circuits. This report covers all the topics which I learned in the training and also the observations made and also the technical abilities gained by me during the internship training.

CHAPTER 2

COMPANY PROFILE

2.1 Introduction

Indian Nippon Electricals Limited (INEL) is a major Indian automotive electrical and electronic components manufacturer. The company designs, develops and manufactures ignition systems, electronic control units, motor controllers, sensors and power electronic devices utilized in the automotive sector.

2.2 History and Establishment

Indian Nippon Electricals Limited is a joint venture between Lucas Indian Service Limited and DENSO Corporation (Japan). The company has become a reliable source for automotive electronic products and solutions to many leading automobile manufacturers in India.

2.3 Products Manufactured

INEL manufactures a wide range of automotive electrical and electronic products, including:

- Flywheel Magneto (FWM)
- Regulator Rectifier (RR)
- Ignition Coil
- Capacitor Discharge Ignition (CDI)
- Transistor Controlled Ignition (TCI)
- Electronic Control Unit (ECU)
- Electronic Fuel Injection (EFI) Systems
- EV Controllers

- Sensors
- Instrument Clusters
- DC-DC Converters

These products are extensively used in two-wheelers, three-wheelers, and electric vehicles.

2.4 Major Customers

INEL supplies products to several leading automobile manufacturers, including:

- TVS Motor Company
- Hero MotoCorp
- Honda Motorcycle & Scooter India
- Yamaha Motor India
- Suzuki Motorcycle India
- Bajaj Auto

The company serves as a Tier-1 supplier to these Original Equipment Manufacturers (OEMs).

2.5 Organizational Structure and Supply Chain

INEL is a well structured organization composed of departments. It comprises different functional areas which work closely to guarantee the best product development, production, testing and customer support. The main functional areas are Product Development, Embedded Software, Production, PCB manufacturing, SMT assembly, THT assembly, Quality Assurance, Testing and Validation, Supply Chain management, Human Resources and Finance.

INEL is a tier-1 supplier in the automotive supply chain. The raw materials and electronic components are purchased from certified suppliers. Those raw

materials are inspected for quality and the components are utilized during production processes. The final products are tested, validated and delivered directly to main automobile constructors.

The normal material flow at INEL is described as:

Suppliers Incoming Inspection Stores Production (SMT/THT) Testing Quality
Inspection Packaging OEM Customers

This ordered supply chain provides high quality and reliability products to customers.

2.6 Quality Standards and Certifications

To ensure product quality and reliability, INEL follows various international standards and certifications, including:

- ISO 9001
- ISO 14001
- ISO 45001
- ISO/IEC 27001
- IATF 16949

These standards demonstrate the company's commitment to quality, safety, environmental management, and continuous improvement.

2.7 Research and Development

At INEL, development of new products and constant innovation within automotive electronics is in a continuously increasing focus. INEL makes R&D investments so that the performance of the product may be improved, the reliability may be raised and both the electrical and conventional vehicles' changing needs can be satisfied.

2.8 Conclusion

Indian Nippon Electricals Limited (INEL), one of the major players in automobile electronics and a dependable Tier-1 supplier to several automobile manufacturers has a strong portfolio of products and has set a strong foothold in the Indian automotive and Electric vehicle industry by its uncompromising quality, state of the art manufacturing & technology capabilities & research driven products.

CHAPTER 3

AUTOMOTIVE E ELECTRONICS PRODUCTS

3.1 Flywheel Magneto (FWM)

One of the key component of the electrical system of two wheeler is the Flywheel magneto (FWM), a machine used to produce the required electrical energy. In FWM, a rotor in the form of a flywheel fitted with permanent magnets rotates about the stator which contains windings made of copper wire. As the engine starts the flywheel also starts rotating at corresponding engine speed. When the magnetic field of the permanent magnets cuts across the stationary stator windings the induced voltage of alternate current type is produced according to the principle of electromagnetic induction.

The electric power produced by the flywheel magneto are used for charging battery, lighting system, ignition system, other electric load on the vehicle. The value of voltage produced is proportional to engine speed and the strength of magnets and also to the number of turn in the stator windings.

Fly wheel magnetos are being used in two wheeler because they are of very simple design, size and low maintenance and also highly dependable and efficient in production of electrical power without any external source. In automobiles it is used as a self energizing system. The generated AC output is then sent to a regulator rectifier where it is converted to a regulated DC voltage required for

battery charging and all the electrical devices of the vehicle. Performance of the flywheel magneto is highly dependent upon the efficiency of the electrical system.

The study on the construction, working and applications of flywheel magneto was conducted during the period of internship at Indian Nippon Electricals Limited.



Figure 3.1: Flywheel Magneto Assembly



Figure 3.2: Flywheel Magneto

3.2 Ignition Coil

An ignition coil is an important part of an ignition system in a spark ignition engine. It transfers the battery's low voltage current into the high voltage needed for an ignition spark between the spark plug electrodes. A step-up transformer, based on the principle of electromagnetic induction is how the ignition coil functions. It's actually two coils on a magnetic core called the primary coil and the secondary coil. When current flows through the primary coil it creates a magnetic field, when the current is cut off it collapses extremely rapidly. A large voltage is induced in the secondary coil, it is this large voltage (anything between several thousand to millions of volts) which is sent to the spark plug so it can spark and combust the fuel mixture. The effectiveness of an ignition coil will determine a engines performance, efficiency and its emissions. The latest ignition systems have been developed to have far more electronic control over when exactly the spark plug sparks at the optimum point. The different types of ignition coils currently being used in vehicles was investigated during the internship.



Figure 3.3: Ignition Coil

3.3 Transistor Controlled Ignition (TCI)

Transistor Controlled Ignition (TCI) is an electronic ignition system in which the flow of current to the ignition coil is regulated by a transistor which is switched ON and OFF. When the transistor switches OFF, a high voltage is induced in the ignition coil which is passed to the spark plug. The TCI system provides accurate ignition timing, better fuel economy and lower emission and increased engine power output. TCI is now commonly found on many two and four wheelers.

3.4 Capacitor Discharge Ignition (CDI)

CDI (Capacitor Discharge Ignition) uses the capacitor to store energy then quickly discharge that energy through the ignition coil producing a high voltage spark. A CDI system charges the capacitor with voltage from the flywheel magneto then upon receiving a pulse from the pulser coil the capacitor is discharge creating a spark at the spark plug. CDI systems offer quick response, reliable ignition and are efficient at high engine speeds, which makes them an ideal system for motorcycles and small engines.

3.5 Integrated Starter Generator (ISG) Controller

The Integrated Starter Generator (ISG) Controller is an advanced electronic system used in vehicles, which integrates the functionalities of the starter motor and the alternator into one device. It can start the engine, provide electrical energy and charge the battery as the vehicle is in operation. The ISG controller governs the flow of power from the battery to the motor-generator set. Power is supplied by the battery to the ISG controller and to the motor while the engine is being started. Once the engine begins operating, the same device begins acting as a generator and supplies power to the electrical consumers while the battery is charged. The ISG technology is highly fuel efficient, emission reduced and enables the start-stop system on vehicles. This internship has helped in gaining

knowledge about the function and uses of ISG controllers in the context of automotive electrical systems, and to obtain an insight into modern fuel efficient vehicles technology and sophisticated power management systems in automobiles.



Figure 3.4: ISG Controller

3.6 Pulser Coil

The pulser coil is an ignition system sensor that tells the ignition system when to fire the spark. The pulser coil creates a tiny pulse of electricity at specific points on the rotating flywheel which is used by the CDI or TCI system to begin the ignition process.

Electromagnetic induction is how the pulser coil functions. As the flywheel rotates it turns an attached permanent magnet. As the magnet passes the pulser coil a voltage pulse will be induced into the coil's winding because of the change in magnetic flux. The pulser coil's output signal is an extremely critical timing reference for the ignition controller. If there is a slight error in timing even this small timing error can cause the engine to not run correctly and use excess fuel. Pulser coils are small, reliable and able to endure difficult automotive environments.



Figure 3.5:Pulser Coil

3.7 Electronic Control Unit (ECU)

The Electronic Control Unit (ECU) is the vehicle's computer system that controls the operations like fuel injection, ignition timing and emissions. The ECU receives signals from various sensors, processes the information and controls actuators. The ECU can provide better vehicle performance, better fuel efficiency and improved reliability.

3.8 Electric Vehicle (EV) Controllers

Electric Vehicle(EV) Controller is a sort of electronic devices which are employed to control and manage the operation of electrical motor for the EVs. It bridges the interface between battery and motor. Control amount of electric power is supplied according to the acceleration as well as condition of vehicle. It processes the information transmitted from brake pedal, accelerator, sensors to adjust the motor velocity, torque, direction etc. EV controller can protect the over current, over voltage, temperature, or abnormal condition during operation of EVs. The modern EV controller has already incorporated high efficient MOSFET and intelligent algorithm to ensure the stability. The market of electric vehicles

has boosted the importance of controller, by interning it, we learn about its basic working process and application in electric vehicles.

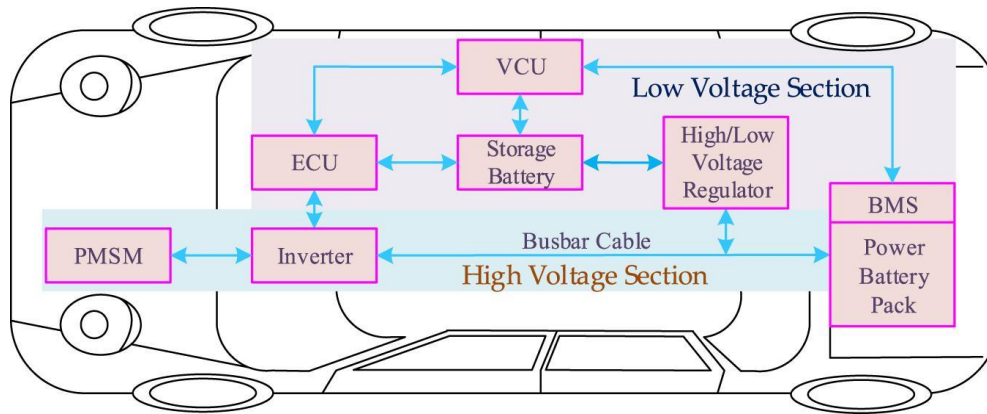


Figure 3.6: Electric Vehicle Controller

CHAPTER 4

PCB DESIGN AND MANUFACTURING PROCESSES

4.1 Introduction to Printed Circuit Board (PCB)

The foundation of any modern electronics is the Printed Circuit Board (PCB). PCB provides mechanical support to the electronic components and it allows the interconnections among the electronic components. In the case of automotive electronics, PCBs are largely used in devices like Regulator Rectifiers (RR), CDI Units, TCI Units, ECU, EV Controllers, Instrument Clusters and DCDC converters.

In a PCB, conventional wirings are replaced by the combination of copper traces, pads, vias, silkscreen, solder mask etc. This kind of wiring increases the reliability and decrease the size of the circuits and also enhance the manufacturability of

circuits. Depending upon the type of application, PCBs can be single layer, double layer or multilayer.

For automotive electronics PCBs developed at Indian Nippon Electricals Limited (INEL) are designed considering the harsh operating environment like vibration, temperature variations, humidity and electrical noises etc. Hence it should be ensured for high reliable products.

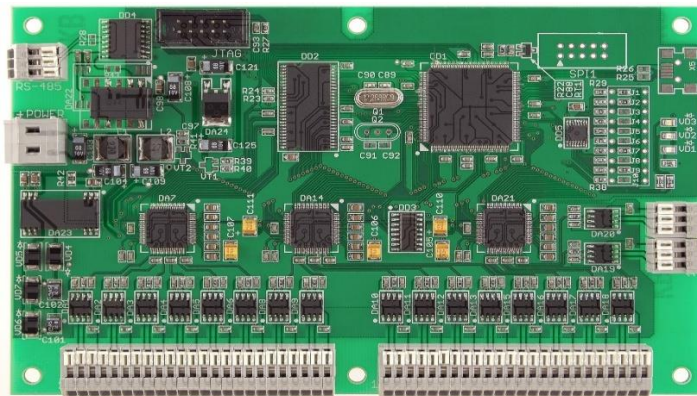


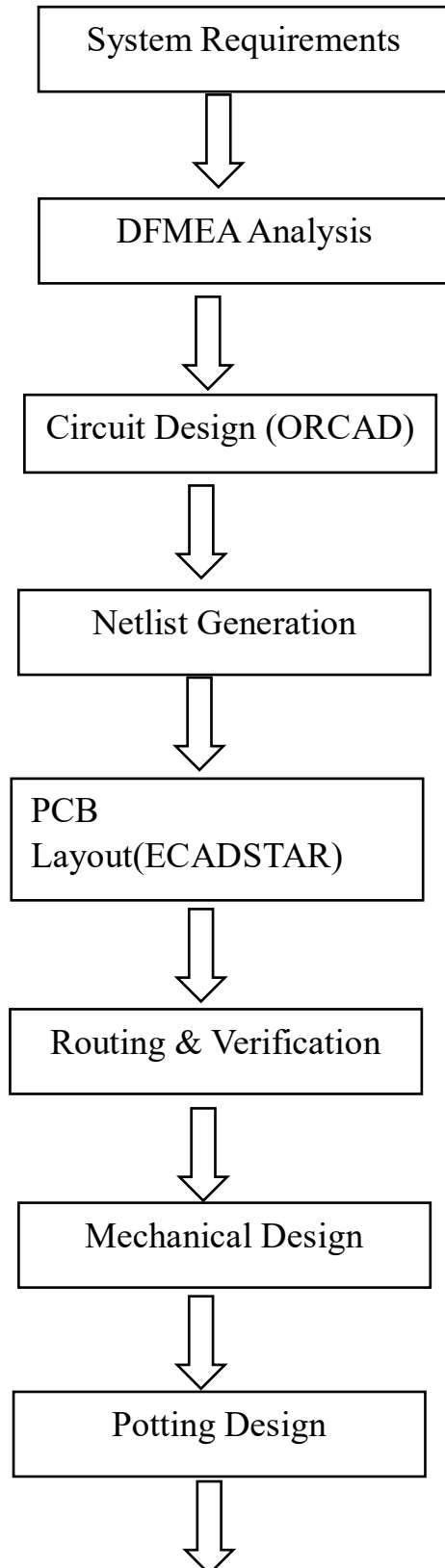
Figure 4.1: Printed Circuit Board

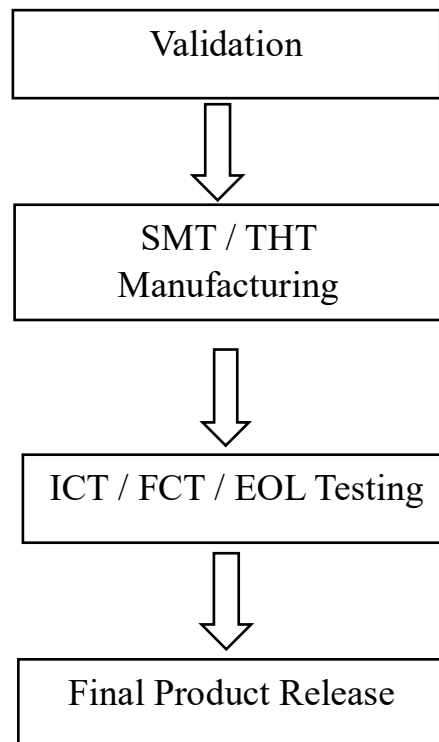
4.2 PCB Design Flow

To start the process for PCB design is to Gather all customer requirements and system requirements. After gathering all the requirements, a DFMEA is done so as to determine possible failure modes so that product can be improved to enhance its reliability. Then a circuit schematic is drawn in ORCAD based on the requirements gathered.

After the schematic is done, the net list is extracted so as to record all the electrical connection between components. Then it is imported to ECADSTAR and PCB layout is designed. In this phase components are placed on the PCB, routed, pads are created and the design is verified.

After finishing the layout, we check on the mechanical design and potting specifications. We then verify the design and release it to be manufactured. Lastly, SMT/THT assembly is done and PCB is then tested.





4.3 PCB Design Software Used at INEL

During the internship, exposure was gained to various PCB design software tools used in product development activities. These tools help engineers convert circuit concepts into manufacturable PCB designs.

4.3.1 ORCAD

ORCAD is used for schematic capture and circuit design. Engineers create electrical schematics containing all electronic components and their interconnections. ORCAD also performs design verification and generates the netlist required for PCB layout development.

The software provides extensive component libraries and helps reduce design errors during circuit development. The schematic developed in ORCAD acts as the foundation for the entire PCB design process.

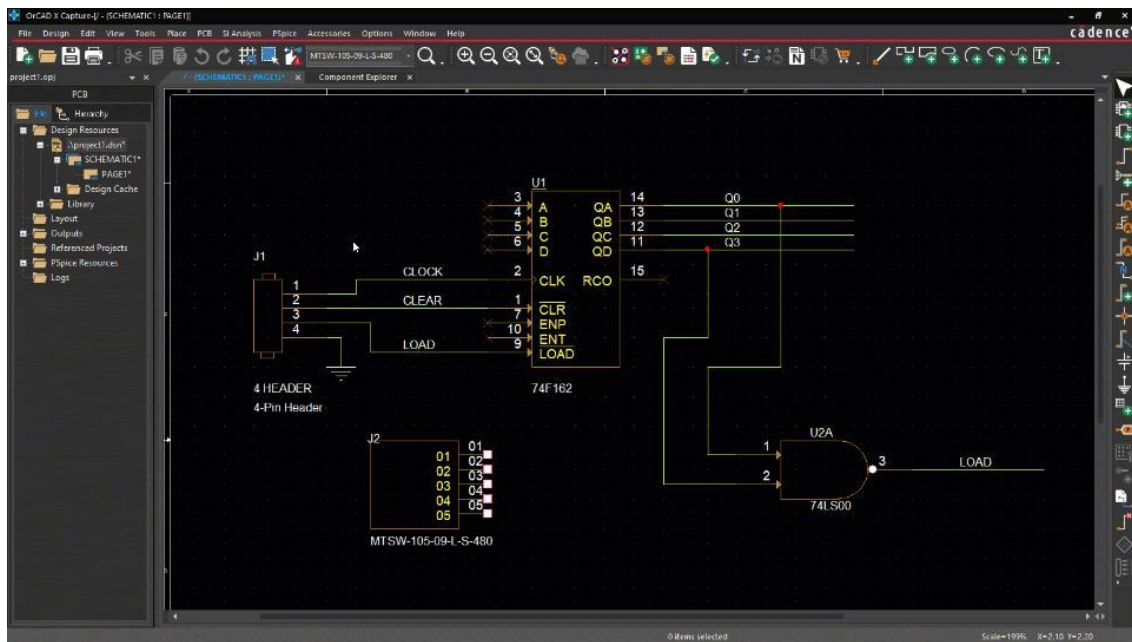


Figure 4.2:ORCAD Software

4.3.2 Netlist Generation

Once the schematic is completed, a netlist is generated. The netlist contains information regarding all electrical connections between the components used in the circuit.

The netlist acts as a communication bridge between the schematic design and PCB layout stages. Any error in the netlist may result in incorrect PCB connectivity. Therefore, proper verification is performed before proceeding to PCB layout development.

4.3.3 ECADSTAR

ECADSTAR is used to design PCB layout. Once the netlist is loaded, the components can be placed on the PCB and copper tracks can be routed in order to make electrical connection.

Multi-layer PCB design, automatic routing, design rule checking and generation of manufacturing files can be performed. Circuit performances are enhanced and EM interferences are reduced when components are well placed and routed.

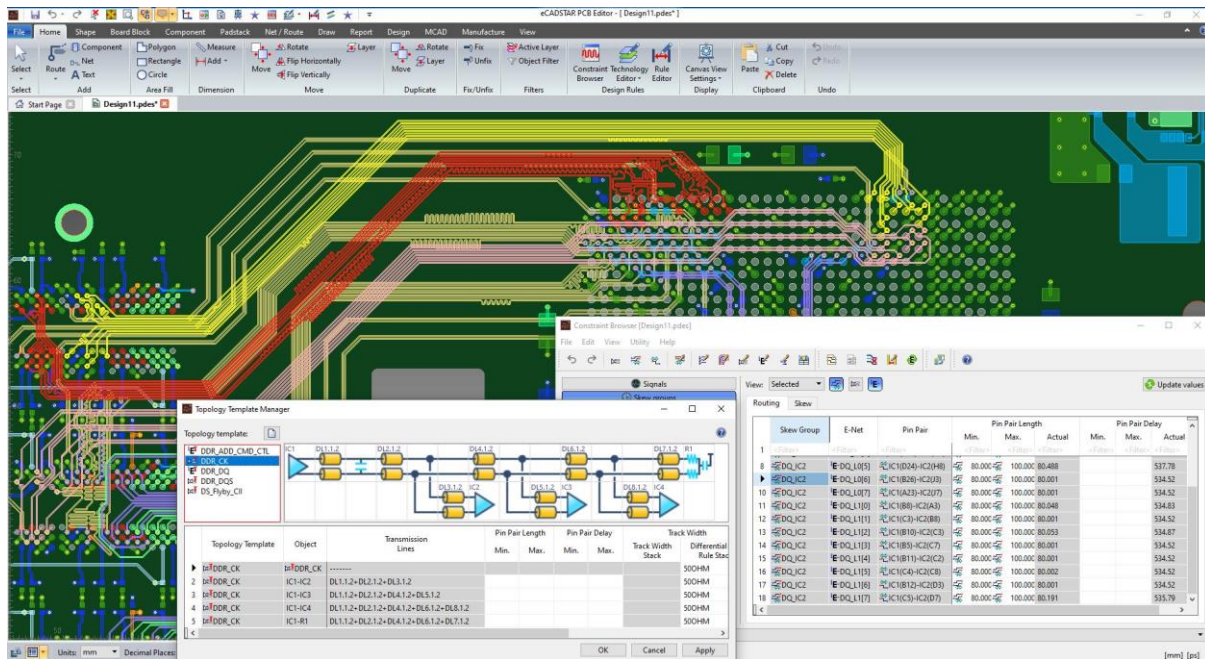


Figure 4.3:ECADSTAR Software

4.3.4 Allegro PCB Designer

Allegro PCB Designer is a high-performance PCB design software utilized to create elaborate electronic devices for automotive and industrial applications. It offers robust capabilities for multi-layer routing, signal integrity checks, design rule verification, and generation of manufacturing files.

Allegro is particularly handy when dealing with automotive electronics which are characterized by small form factor designs with high reliability and strict design regulations.

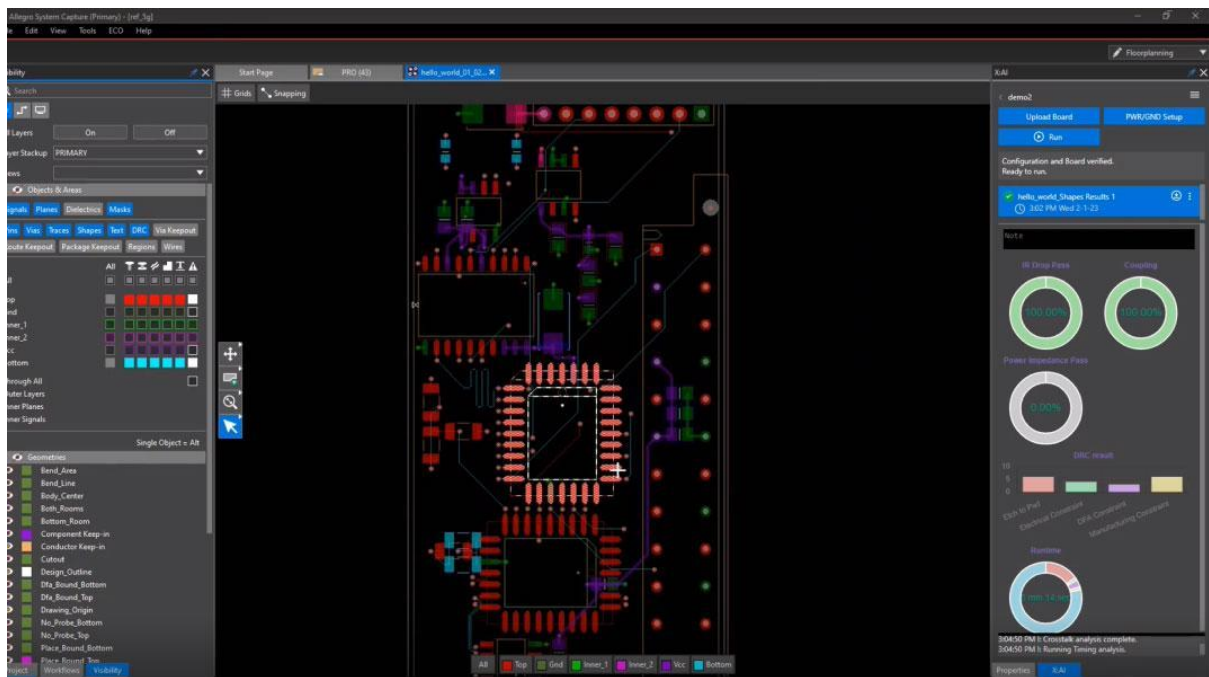


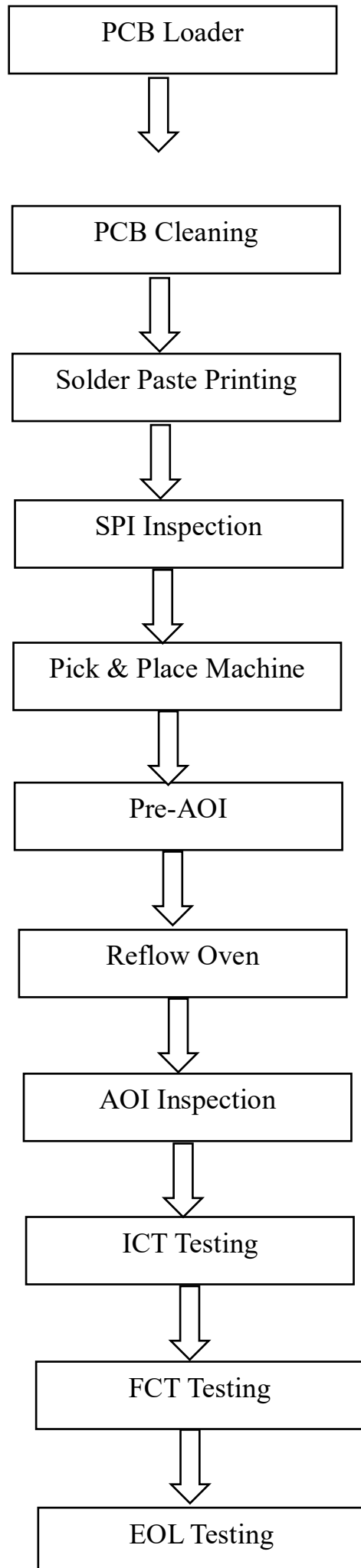
Figure 4.4:Allegro Software

4.4 Surface Mount Technology (SMT)

Surface Mount Technology (SMT) is a method for mounting electronic components onto the surface of a PCB. The use of SMT allows smaller circuit design, greater component density, decreased manufacturing cost and higher electrical performance. SMT assembly is used extensively for automotive electronic products at INEL, as it provides speed, accuracy and reliability. The SMT assembly includes the subsequent steps: paste printing, pick and place, reflow, inspection, test.

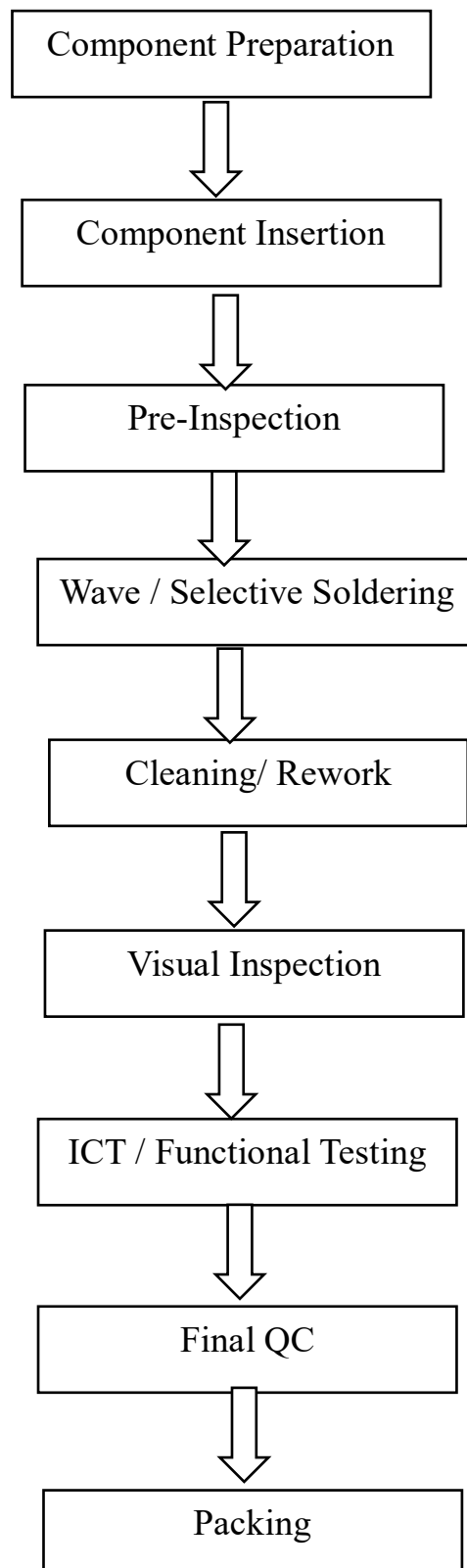


Figure 4.5: SMT



4.5 Through Hole Technology (THT)

Through Hole Technology (THT) is a PCB assembly technique where component leads are inserted through drilled holes and soldered on the opposite side of the PCB. THT provides strong mechanical support and is commonly used for connectors, transformers, relays, and power components.



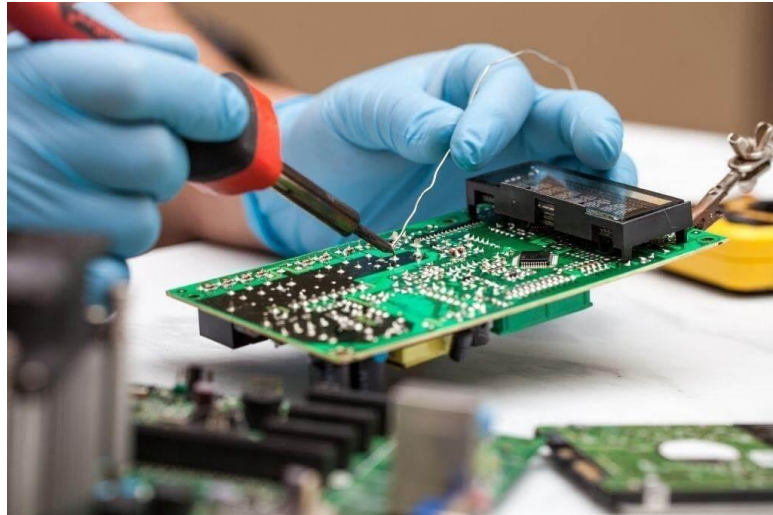


Figure 4.6: THT

4.6 PCB Types and Vias

There are different classifications for printed circuit boards depending on how many conductive layers the board uses to form connections. There are three main types of PCBs: Single-layer PCB, Double-layer PCB, and Multi-layer PCB.

Single-layer PCB

In a single-layer PCB there is one conductive layer to place components on and route the circuit through.

Double-layer PCB

In a double-layer PCB there is a conductive layer on top of the board and on the bottom of the board. This allows for more possible routes for signals.

Multi-layer PCB

A multi-layer PCB has several conductive layers stacked on top of one another. This could include a 2-layer board, 4-layer board, 6-layer board or an 8-layer board etc, but will generally always be in an even number.

For example the layout of a standard 4-layer board will generally be arranged in this way.

Layer 1: Components and signal rout.

Layer 2: Ground plane.

Layer 3: Power plane.

Layer 4: Components and signal rout.

Multilayer PCBs are often beneficial when it comes to reducing noise and improving the performance of a circuit.

PCB Measurement Units

Measurements on PCBs are generally referred to in both mm (millimetres) and mils.

1 mm = 40 mils

4 mil = 0.1 mm

These units are used when specifying the width and spacing of traces and any other parameters of the PCB layout.

Types of Vias

A via is essentially a hole that is drilled into the PCB, enabling a connection between layers to be formed. There are three different types of vias:

Blind Via

A blind via goes from an outside layer, into some of the inside layers of the PCB without coming out the other side of the board.

Through-Hole Via

A through-hole via runs from the very top of the PCB to the very bottom of the PCB, therefore enabling connection between all layers.

Buried Via

A buried via is the opposite to a blind via, in that it enables connection between two inner layers of the board, not the outside ones.

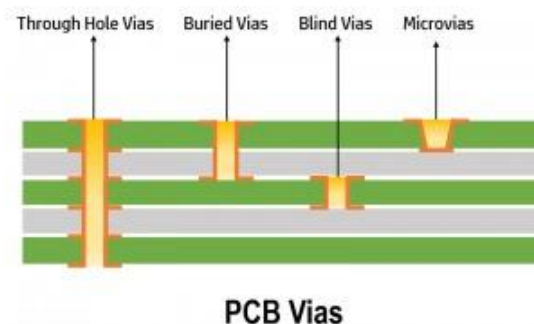


Figure 4.7:PCB vias

CHAPTER 5

MOTOR CONTROL SYSTEMS

5.1 Introduction to Electric Motors

Electric motors are electromechanical transducers converting electricity into mechanical energy. They are used in all the electric vehicles, automation, domestic appliances, automobiles and etc. Functioning principle of electric motors depends upon interaction between electric current and magnetic fields. An electric motor in an electric vehicle is intended for providing necessary torque and movement.

5.2 Types of Motors Used in Electric Vehicles : According to the training provided at INEL, two major types of motors are commonly used in electric vehicle applications:

Permanent Magnet Synchronous Motor (PMSM)

In PMSM motors the rotor contains magnets, and the stator contains the 3 phase windings. These motors are highly efficient, have a high torque density and have excellent speed regulation. PMSM motors are increasingly used in modern electric vehicles due to their high efficiency and performance. The EV Controller at INEL controls the PMSM motor, and these use a brushless type of PMSM motor.



Figure 5.1: PMSM Motor

Induction Motor

This type of motor works by using electro-magnetic induction principles. PMSM motors differ in that induction motors use no permanent magnets in their rotor design and rely on induced rotor currents created by the stator magnetic field to drive the motor. They are generally considered reliable and cost-effective in high power operation.

5.3 Three-Phase Motor Fundamentals

Almost all electric vehicle motors use a three phase input power supply. Let R, Y, B denote the incoming phases of the supply and U, V, W the motor winding terminals. These phases are separated from one another by 120 electrical phase difference. The flow of the three-phase currents in stator windings produces a rotating magnetic field. This magnetic field interacts with the rotor and creates torque which turns the motor shaft.

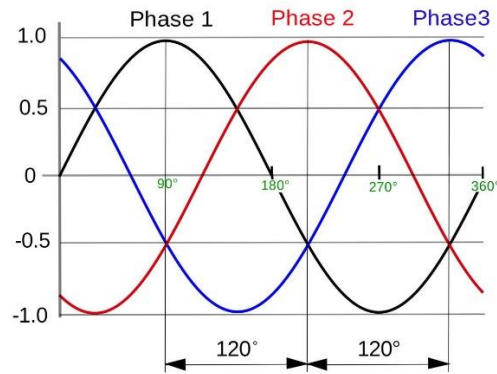


Figure 5.2: AC Motor Waveform

5.4 Rotor and Stator Construction

A motor comprises two main components; rotor and stator. Stator is the stationary component and the windings to produce the magnetic field is located there. While the rotor is the rotating component that is fixed to the motor shaft. Permanent magnets are employed in PMSM motors at the rotor. The magnetic field of permanent magnets is combined with the rotating magnetic field of the stator and hence rotation of the rotor occurs by electrical to mechanical energy conversion.

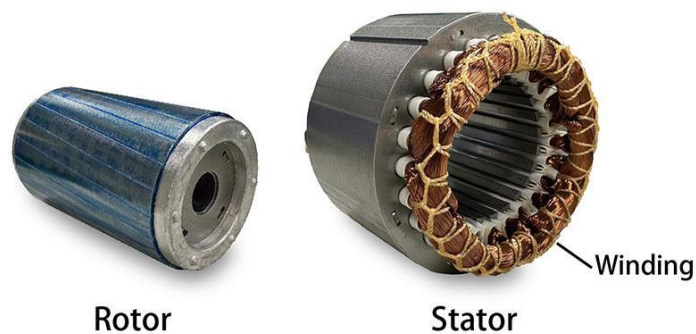


Figure 5.3: Rotor and Stator

5.5 Motor Windings

Motor windings are arranged inside the stator slots to generate the magnetic field required for motor operation.

Lap Winding : In lap winding, the coil ends are connected to adjacent commutator segments. This arrangement provides multiple parallel paths for current flow and is suitable for high-current applications.

Wave Winding : In wave winding, the coil connections progress in a wave-like pattern around the armature. This winding provides fewer parallel paths and is suitable for higher voltage applications.

5.6 DC Link Capacitor

The DC Link Capacitor, one of the key components of an EV controller, is situated between the battery and inverter stage across the DC bus. Its main function is to minimize the ripple voltage in order to ensure a smooth dc voltage. This energy-storage device, absorbing the current shock due to the load variation during accelerating, braking and sudden load variation, can ensure the performance of the inverter and protecting electronic components.

5.7 MOSFET Power Stage

In the EV controller, MOSFETs have been used as high speed switches. MOSFET converts DC input from battery into a regulated 3 phase AC signal that will be fed to the motor.

The configuration consists of upper and lower switching arms made of MOSFET. The three phase inverter that drives U, V and W motor windings consist of several MOSFETs together. Their sequence of switching is governed by PWM generated by the controller.

5.8 Gate Driver and Bootstrap Circuit

A gate driver is employed to give the required gate voltage for the MOSFET to turn ON. For high-side MOSFET switching, the gate voltage must be higher than the supply voltage, so a bootstrap circuit is used. The bootstrap circuit includes bootstrap capacitor, resistance and diode. It stores energy and provides the extra

voltage that is necessary to turn ON high-side MOSFET, hence operates the inverter and control motor efficiently.

5.9 Brushless DC Motor (BLDC)

BLDC Motor is a brushless D C motor, which is electronically commutated. The stator windings are excited using electronic switching circuits depending on the position of the rotor. BLDC motors have advantages such as high efficiency, low maintenance, long life and good speed control, etc., and these advantages have led them to be used for electrical vehicles, industrial drive systems and automation.

5.10 Working of EV Controller

The controller is the brains of the EV drive system. It receives information from the throttle, sensors, battery pack, motor feedback circuits. Based on the input signals received the controller produces the PWM signals for the MOSFET inverter.

The inverter converts DC power from the battery into a switched 3-phase A/C output to the motor. The controller monitors the speed, current, voltage and temperature to insure safe operation of the vehicle.

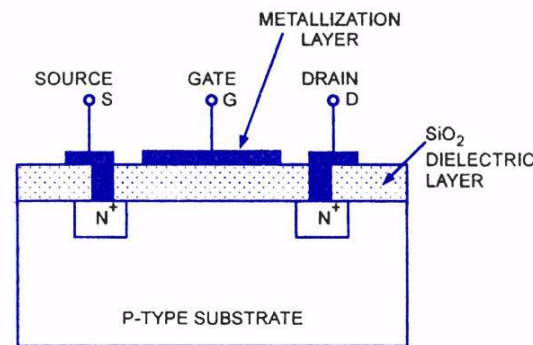
CHAPTER 6

MOSFETS AND POWER ELECTRONIC CIRCUITS

6.1 Introduction to MOSFET

MOSFET (Metal Oxide Semiconductor Field Effect Transistor) is a type of semiconductor device, which is mostly employed as an electronic switch in power electronics circuits. MOSFET is the power device to switch electric power on/off with high efficiency and it is often used in EV controllers, DC-DC converters, motor drives, automotive electronic system, etc. High switching speed, less

power dissipation, miniaturization are desirable characteristics of MOSFET in modern automotive design.



N-Channel E-MOSFET Structure

Figure 6.1: MOSFET Structure

6.2 Working Principle of MOSFET

A MOSFET is a three terminal device and the three terminals are G(gate), D(drain), and S(source). When a voltage is applied between G and S, the electric field created switches off or on the current which is allowed to flow between D and S. When the gate voltage is above the value of V_T the MOSFET is on and the current flows between the drain and the source, if no gate voltage is applied the device will be off and no current will flow.

6.3 Advantages of MOSFET over Relay

Switching is carried out through the use of mechanical contacts in relays but through an electronic process by a MOSFET. MOSFET's have the advantages of operating at higher speeds, low noise, zero maintenance and a much longer lifespan over relays. MOSFET's are also smaller and more robust than relays and are ideal for high-speed switching applications, and thus for the application in automotive electronic control systems and EVs controllers.

6.4 MOSFET Parameters

The selection of a MOSFET depends on several important parameters.

Drain-to-Source Voltage (VDS) : VDS is the maximum voltage that can be applied between the drain and source terminals without damaging the MOSFET.

Drain Current (ID) : ID represents the maximum current that can safely flow through the MOSFET.

RDS(ON) : RDS(ON) is the resistance between drain and source when the MOSFET is fully turned ON. Lower RDS(ON) results in lower power loss and higher efficiency.

Gate Charge (Qg) : Gate charge determines how quickly the MOSFET can switch ON and OFF.

Junction Temperature : This parameter indicates the maximum temperature the MOSFET can withstand during operation.

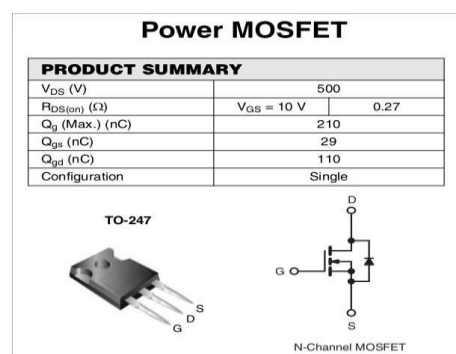


Figure 6.2: MOSFET parameters

6.5 MOSFET Selection for Automotive Applications

When selecting MOSFETs for automotive systems, choices are guided by battery voltage, current needs, switching frequency and thermal and safety considerations. For 12V automotive systems, the majority of devices used have a voltage rating around 30-60V. In high voltage EV systems, very high voltage rated MOSFETs are chosen. thermal and safe operating area factors should also be taken into consideration.

6.6 Thermal Considerations

Heat is produced in MOSFETs during their operation through both conduction and switching losses. High temperatures can significantly decrease efficiency and cause the device to fail.

To prevent this heat from building up, heat sinks, thermal pads and appropriate PCB design strategies are employed to extract the heat. Thermal calculations are undertaken to identify and ensure the appropriate MOSFET is chosen to operate reliably under all the conditions.

6.7 Safe Operating Area (SOA)

The SOA provides the limits of voltage and current a MOSFET can work at without failure; above these limits the MOSFET will not survive.

Voltage transients and voltage spikes can happen quite often in automotive systems and so it is common practice to try and design the MOSFET to operate within the limits of the SOA under both fault and non-fault conditions.

6.8 Gate Driver Compatibility

The gate driver circuits supply the necessary voltage and current to enable the MOSFET to switch effectively. The gate driver needs to be able to supply adequate gate voltage and current for fast switching, hence ensuring minimum loss. By selecting the correct gate driver circuits, the switching characteristics of the system could be enhanced leading to better efficiency. The automotive motor controllers are dependent on well-designed gate driver circuits.

6.9 Applications of MOSFETs in Automotive Electronics

They can be utilized in EV controllers, DC-DC converters, battery management system, motor drivers, lighting, and automobile control units. The MOSFET is an ideal choice in automobile applications owing to the switching speed, high efficiency and miniaturization.

CHAPTER 7

DC-DC CONVERTERS

7.1 Introduction to DC-DC Converters

A DC-DC converter is an electronic power circuit which converts one DC voltage into another DC voltage level. In vehicles they are used for powering numerous systems with a regulated voltage. Their benefit are improved efficiency and output voltages regulation.

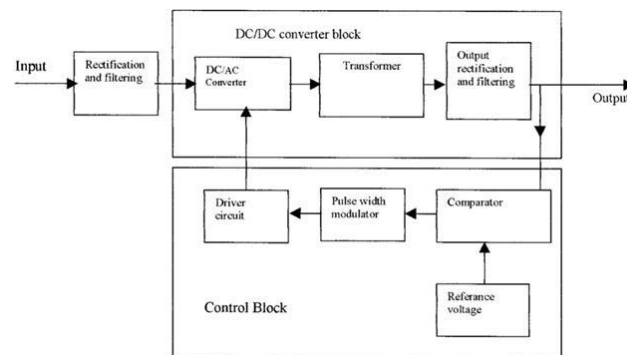


Figure 7.1: Converter Block

7.2 Need for DC-DC Conversion

In automotive systems, the battery voltage can fluctuate depending on the charge and discharge of the battery. All electronic circuits require a stable voltage to function correctly, thus DC-DC converters are used to maintain the correct voltage and improve system efficiency. For example, the converter might be used to transform fluctuating battery voltages of 36-58V into a steady output of 14V for automotive electronics.

7.3 Classification of DC-DC Converters

DC-DC converters are classified into:

Non-Isolated Converters

- Buck Converter

- Boost Converter
- Buck-Boost Converter
- CUK Converter
- SEPIC Converter

Isolated Converters

- Flyback Converter
- Forward Converter
- Push-Pull Converter
- Half Bridge Converter
- Full Bridge Converter

7.4 Buck Converter

A Buck converter is a non-isolated type DC to DC converter, which is used to convert high voltage DC to low voltage DC. One of the most widely used converter topologies in power electronics, because of the simple construction, efficient working and reliable operation. In Indian Nippon Electricals Limited (INEL), the Buck Converter topology is used to produce regulated output voltage for automotive electronic systems by making it in DC to DC converter. It is mainly consists of a MOSFET switch, inductor, capacitor, diode, and load. The output voltage can be obtained at a low voltage level compared to input by the control of the MOSFET switching action.

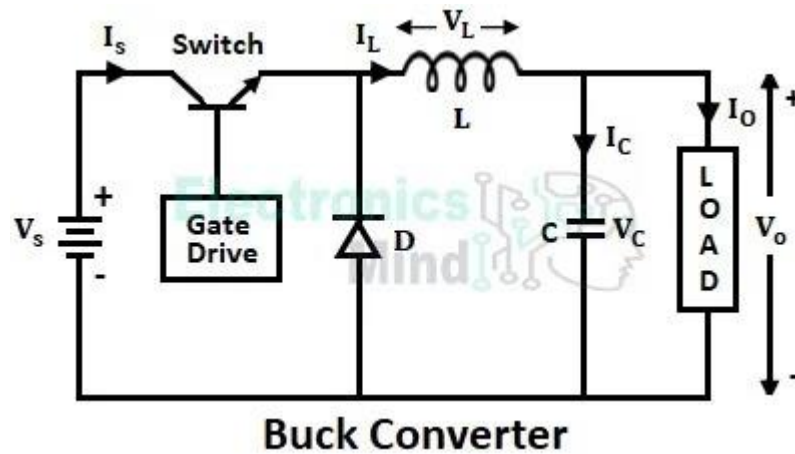


Figure 7.2: Buck Converter

7.4.1 Components of Buck Converter : The Buck Converter consists of the following major components:

MOSFET : Acts as a high-speed electronic switch and controls the energy supplied to the load.

Inductor (L) : Stores energy when the MOSFET is ON and releases energy when the MOSFET is OFF.

Capacitor (C) : Reduces output voltage ripple and maintains a smooth DC output voltage.

Diode : Provides a path for inductor current when the MOSFET is switched OFF.

Load : Consumes the output power supplied by the converter.

7.4.2 Working Principle of Buck Converter

The operation of the Buck Converter can be understood in two modes.

Mode 1: MOSFET ON Condition

When the MOSFET is switched ON:

- Input voltage is applied across the inductor and load.
- Current flows through the MOSFET, inductor, capacitor, and load.
- The inductor stores energy in the form of a magnetic field.
- The capacitor gets charged and supports the output voltage.
- The diode remains reverse biased.

Mode 2: MOSFET OFF Condition

When the MOSFET is switched OFF:

- The inductor opposes the sudden change in current.
- The stored magnetic energy in the inductor starts releasing.
- Current flows through the diode and load.
- The capacitor helps maintain a constant output voltage.
- Energy stored in the inductor continues supplying the load.

This ON and OFF switching process occurs repeatedly at high frequency, resulting in a lower average output voltage than the input voltage.

7.4.3 Output Voltage Control

The output voltage of a Buck Converter depends on the duty cycle of the MOSFET switching signal.

- Higher duty cycle → Higher output voltage
- Lower duty cycle → Lower output voltage

The relationship is:

$$\mathbf{V_{out} = D \times V_{in}}$$

where:

- V_{out} = Output Voltage
- V_{in} = Input Voltage
- D = Duty Cycle

By changing the duty cycle, the converter can regulate the output voltage according to the application requirements.

7.4.4 Types of Buck Converters

Buck converters are mainly classified into two types:

A. Non-Synchronous Buck Converter

The non-synchronous buck converter use a MOSFET as the switch and a diode as the freewheeling circuit. When the MOSFET is switch ON, it conduct current and provides energy for the load. When the MOSFET is switch OFF, the current flow through the inductor free wheels through the diode. Although the circuit is economical and simple, a power loss will be imposed to the diode due to the forward voltage drop. Therefore the efficiency of non-synchronous buck converter is not too high, especially with large current.

B. Synchronous Buck Converter

The standard diode is replaced by a second MOSFET, which result into two MOSFET's in the circuit, where we used a MOSFET and a diode earlier. As this second MOSFET turned ON when the first MOSFET turned OFF it has provided a low resistance path for the inductor current. The second MOSFET also reduces conduction loss, because resistance of a MOSFET is very small as compare to the voltage drop across a diode, the conduction loss has greatly reduced so, efficiency of synchronous buck converter is very high from standard buck converter and this is now very popular for Automotive and EVs applications.

7.5 Boost Converter

Boost Converter: A boost converter is a step-up converter where the input voltage is stepped up to provide higher output voltage. The boost converter utilizes the inductor to store energy and then the inductor transfers this stored energy in to output and hence the output voltage becomes higher than the input voltage.

Battery driven systems, renewable energy systems etc employ boost converter.

7.6 Buck-Boost Converter

A Buck-Boost Converter can both increase and decrease the input voltage depending on operating conditions. It combines the characteristics of buck and boost converters and is useful when the input voltage varies above and below the desired output voltage.

7.7 SEPIC Converter

SEPIC (Single Ended Primary Inductor Converter) is a non-isolated converter that can achieve an output voltage that is higher or lower than input voltage with positive polarity at the output.

Commonly used in the automotive power supply where battery voltage can range wide.

7.8 CUK Converter

The CUK converter is a non-isolated D C to D C converter which can be step-down and step-up. It employs two inductors, a capacitor, a MOSFET and a diode to transfer the energy. The interesting characteristic of the CUK converter is that the polarity of the output voltage is inverse to that of input voltage. It has the advantage of continuous input and output current, small ripples and hence can be

used in power electronics and automobile fields. However, the circuit complexity is higher than buck and boost converters.

7.9 Protection Features in DC-DC Converters and EV Controllers

7.9.1 Over Voltage Protection (OVP)

Over Voltage Protection: This protects electronic components from damage by restricting or terminating the circuit upon detection of an over-voltage (too high an input or output voltage).

7.9.2 Under Voltage Protection (UVP)

Under Voltage Protection safeguards the system against instances where the supply voltage may be less than necessary to operate properly.

7.9.3 Over Current Protection (OCP)

Over Current Protection-This protection mechanism safeguards the circuit from the over flow of current when there is overload or fault in the system. On crossing the pre-defined current value, the controller drops or cuts the flow of current.

7.9.4 Short Circuit Protection (SCP)

Short Circuit Protection is designed to protect the power electronic devices from excessively high currents which may result in a short circuit. The controller rapidly turns the circuit OFF so that the components are not damaged.

7.9.5 Reverse Polarity Protection

Reverse Polarity Protection allows the unit to not be damaged due to incorrect battery terminal hook-up. It is also designed to prohibit current flow in the reverse direction and stop failure of sensitive electronic components.

7.9.6 Thermal Protection

Thermal Protection is used to monitor the temperature of the power devices such as MOSFETs and controllers. When the power device has a temperature greater than safe operation conditions it can reduce the power or shut down the device to avoid damage.

7.10 Isolated DC-DC Converters : Isolated DC-DC converters use transformers to provide electrical isolation, ensuring improved safety and high-voltage operation. Common types include flyback, forward, push-pull, half-bridge, and full-bridge converters used in industrial and automotive applications.

CHAPTER 8

LTSPICE SIMULATION OF ZERO CROSSING DETECTOR

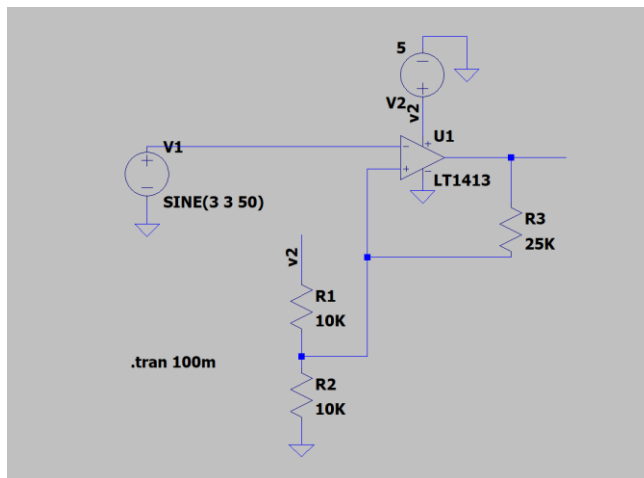
8.1 Introduction to LTspice

LTspice is a simulation software widely used for designing and analyzing electronic circuits. It enables engineers to verify circuit performance before hardware implementation.

8.2 Zero Crossing Detector

A Zero Crossing Detector is a comparator circuit used to detect when an input signal crosses zero volts. It converts a sinusoidal waveform into a square wave output.

8.3 Circuit Design



8.4 Working Principle of Zero Crossing Detector

The Zero Crossing Detector circuit illustrated above is configured with a LT1413 operational amplifier used as a comparator with positive feedback. An AC sine wave V1 (SINE(3 3 50)) is the input to the circuit, and R1, R2, R3 represent the positive feedback network that contributes to the hysteresis of the circuit to prevent the rapid switching of the output from flickering.

When the input signal is higher than the reference threshold voltage, the output of the comparator switches to high (roughly 5V). When the input signal drops below the reference threshold voltage, the output of the comparator switches to low (roughly 0V). Therefore the input sine wave is converted to a square wave.

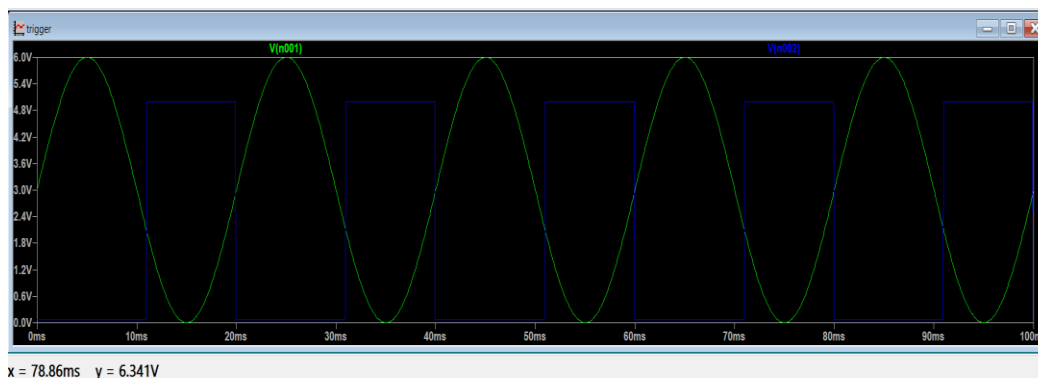
Resistors R1, R2 form the voltage divider that creates the reference voltage. Resistor R3 contributes positive feedback to the system. This positive feedback makes the circuit's output not change states if the input is too close to the threshold voltage to provide a stable output and keep the output from changing values in an unwanted way due to noisy input signal.

As the input sine wave passes through the reference voltage twice for each cycle the output changes. For these reasons the zero-crossing detector can be used to detect the crossing points of the sine wave and thus can be used for signaling, converting waveforms, measuring frequency, phase control, and synchronization applications.

8.6 Components Used

- LT1413 Operational Amplifier
- Input Sinusoidal Source (V1)
- DC Supply Voltage (5 V)
- $R1 = 10\text{ k}\Omega$
- $R2 = 10\text{ k}\Omega$
- $R3 = 25\text{ k}\Omega$

8.5 Simulation Result



8.7 Observation

1. The Zero Crossing Detector successfully converted the sinusoidal input signal into a square wave output by switching its output state whenever the input crossed the reference level.

2. The positive feedback provided by resistor **R3** introduced hysteresis, resulting in stable switching and reducing the effect of noise near the crossing point.

CHAPTER 9

LTSPICE SIMULATION OF BUCK CONVERTER

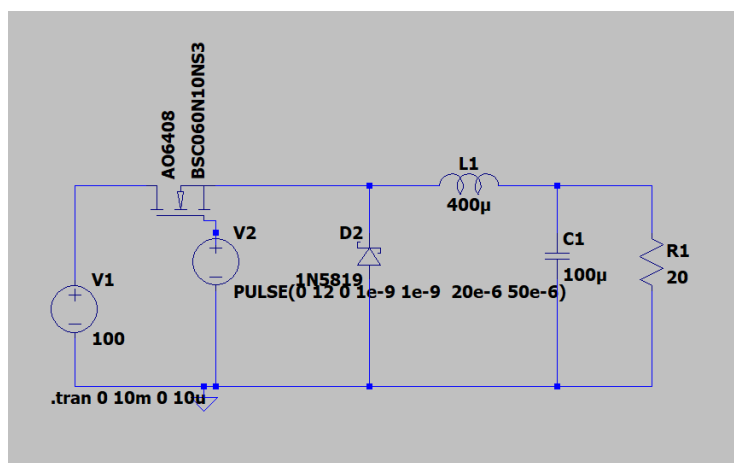
9.1 Introduction

A Buck Converter is a DC-DC power converter used to reduce a higher DC input voltage to a lower DC output voltage. It is widely used in automotive electronics and power supply applications because of its high efficiency and simple design. In this work, a closed-loop Buck Converter was simulated using LTspice to study voltage regulation and converter operation.

9.2 Buck Converter

The Buck Converter is a DC-DC step down converter. This device converts a high DC input voltage to a lower regulated DC output voltage. The duty cycle of the MOSFET control signal is changed in order to regulate the output voltage.

9.3 Circuit Design

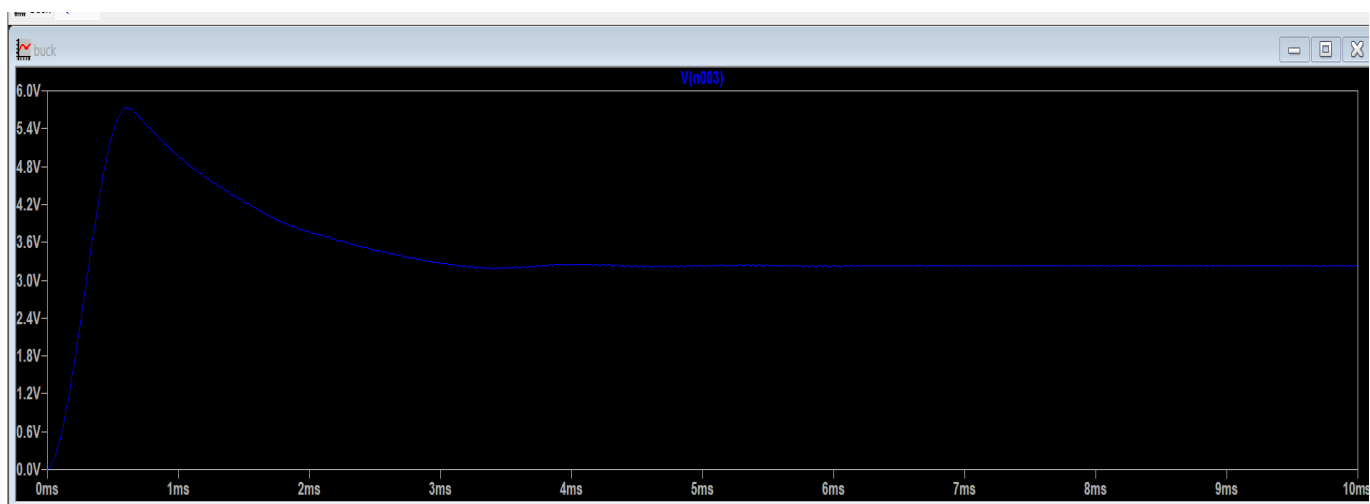


9.4 Working Principle

There are two switching modes which define the operation of the Buck Converter and which is dependent upon the PWM signal applied to the MOSFET. When the MOSFET is switched ON, the input supply is connected to the inductor and current is supplied to the inductor and subsequently to the load. The inductor stores energy during this stage and also the capacitor is charging, supplying power to the load and preventing the output voltage from falling rapidly.

When the MOSFET is switched OFF, the inductor tries to prevent the instantaneous drop in current and the stored energy is then delivered to the load via the diode. The capacitor supplies the load during this stage and the ripple in the output voltage is reduced. By continually alternating between these two modes of switching, a DC output voltage less than the input voltage is obtained and the output level is dictated by the duty cycle of the PWM signal.

9.5 Simulation Result



9.6 Components Used

- **MOSFET (M1):** Acts as the switching device.

- **Diode (1N5819):** Provides a current path when the MOSFET is OFF.
- **Inductor ($L1 = 400 \mu\text{H}$):** Stores and releases energy.
- **Capacitor ($C1 = 100 \mu\text{F}$):** Reduces output voltage ripple.
- **Load Resistor ($R1 = 20 \Omega$):** Represents the output load.
- **Operational Amplifiers ($U1, U2$):** Used for feedback and voltage regulation.
- **Resistors ($R2, R3, R4$):** Form the feedback network.
- **Pulse Source ($V2$):** Generates the switching PWM signal.

9.7 Observation

1. The Buck Converter successfully reduced the 100 V input voltage to a lower regulated output voltage.
2. The feedback circuit maintained a stable output voltage by controlling the switching operation of the MOSFET, demonstrating closed-loop voltage regulation.

CHAPTER 10

FUNDAMENTALS OF AUTOMOTIVE ENGINE

10.1 Introduction to Internal Combustion Engine

An Internal Combustion (IC) Engine converts the chemical energy of fuel into mechanical energy through combustion inside the engine cylinder.

10.2 Four-Stroke Engine

The four-stroke engine completes one cycle in four piston strokes and two crankshaft revolutions.

Intake Stroke : Air-fuel mixture enters the cylinder.

Compression Stroke : The mixture is compressed.

Power Stroke : Combustion pushes the piston downward.

Exhaust Stroke : Burnt gases are expelled.

10.3 Top Dead Centre (TDC)

TDC is the highest position reached by the piston inside the cylinder.

10.4 Bottom Dead Centre (BDC)

BDC is the lowest position reached by the piston during operation.

10.5 Ignition Timing

Ignition timing refers to the instant at which the spark plug generates a spark to ignite the air-fuel mixture.

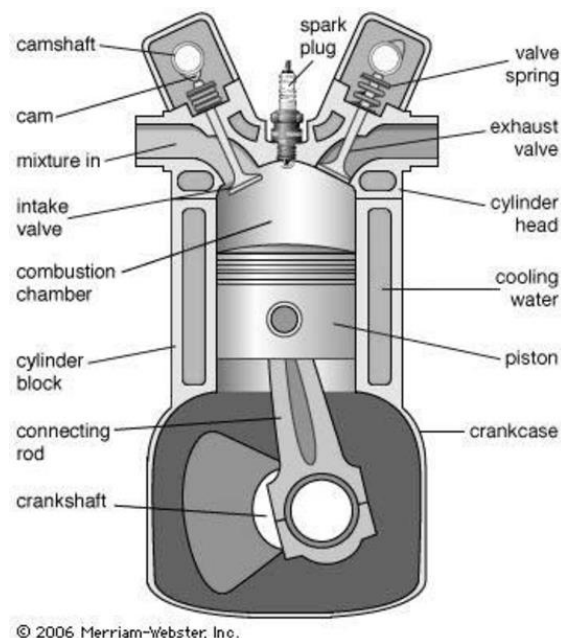


Figure 10.1: Internal Structure of IC engine

CHAPTER 11

SENSORS AND SENSOR APPLICATIONS IN AUTOMOTIVE SYSTEMS

11.1 Introduction

Sensors are an important part of automotive systems nowadays, it measures a range of physical parameters and translates them into electrical signals which can be processed. These electrical signals are utilized by the control units in a vehicle to enhance the performance, efficiency and safety of the vehicle. During the internship, an understanding on different types of sensors and its use in automobiles was obtained at INEL.

11.2 Sensor

An apparatus which receives the change of physical value (temperature, pressure, speed, position, light, etc.) to its own change of electricity signal and then electronic control unit receives its electricity signal.

11.3 Transducer

A transducer is simply a device that is able to transform one form of energy to another. Transducers are used in cars, sensors, to translate a physical magnitude into an electrical one.

11.4 Hall Effect Sensors

Hall effect sensors are extensively used in automobiles. These sensors work on Hall effect principle and the available types are 1D, 2D, and 3D sensors, depending on the applications. Hall sensors contain a Hall plate, which produces

an output voltage when it comes in proximity with a magnetic field. These sensors do not require a physical contact and have a very high reliability.

Applications of Hall Sensors

- Speed sensing
- Position sensing
- Current sensing
- Proximity detection

11.5 Types of Automotive Sensors

Speed Sensor : The Speed section operates within a voltage range of 4.5 V to 16 V, ensuring compatibility with various systems. It performs reliably in extreme temperatures, from -40°C to +200°C. With a maximum output frequency of 8000 Hz and a current consumption of 9 mA max, it supports high-speed functionality.



Figure 11.1: Speed Sensor

Oil Level Sensor: Measures the oil level within a system using a 4.5V to 18V DC operating voltage. It operates at temperatures between -20°C and +180°C, with an output voltage ranging from 5% to 95% of the supply voltage (Vs). It has minimal error in output (± 0.1 V).



Figure 11.2: Oil Level Sensor

Side Stand Indication Sensor: This sensor detects the position of the side stand in two-wheelers and provides indication to the rider, thereby improving safety.

Throttle Position Sensor (TPS): This sensor operates with a 5V input and draws a current of 20mA. It measures angular position from 0° to 360°, with an output voltage range of 5% to 95% of V_s . It functions in various conditions with minimal pressure requirements.



Figure 11.3: Throttle Position Sensor

Gear Position Sensor: A sensor for determining gear position, operating with a 5V supply and measuring angles up to 360°. It operates within the temperature range of -40°C to +150°C and has a robust IP68 rating, ensuring resistance to dust and water.



Figure 11.4: Gear Position Sensor

Temperature Sensor: The Temperature section is designed to withstand temperatures ranging from -40°C to $+120^{\circ}\text{C}$. It operates with an IP6K7 & IP6K9K rating, providing robust protection in harsh environments. The section has no specified operational frequency or current requirements, making it adaptable in various temperature-related applications.



Figure 11.5: Temperature Sensor

ABS Wheel Speed Sensor: The Speed section operates within a voltage range of 4.5 V to 16 V, ensuring compatibility with various systems. It performs reliably in extreme temperatures, from -40°C to $+200^{\circ}\text{C}$. With a maximum output frequency of 8000 Hz and a current consumption of 9 mA max, it supports high-speed functionality.



Figure 11.6: ABS Wheel Speed Sensor

TMAP Sensor: A sensor that operates with a voltage input of $5V \pm 0.25V$, designed for measuring temperature and pressure in environments ranging from -40°C to $+140^{\circ}\text{C}$. The output voltage varies from 4% V_s to 96% V_s and is used for measuring pressure between 20 Kpa and 450 Kpa.



Figure 11.7: TMAP Sensor

Pressure Sensor - 10 Bar: Provides the temperature and pressure specifics of the sensor mounted location.



Figure 11.8: Pressure Sensor

Reverse Parking Assist System – Sensors: This device operates with a frequency of 55.5 ± 2 kHz and supports an input voltage range from 10V to 36V DC. It has a low consumption current of up to 10 mA and operates efficiently within a temperature range of -40°C to $+85^{\circ}\text{C}$. With an IP6K7 and IP6K9K rating, it is highly durable and protected against dust and water ingress.



Figure 11.9: Reverse parking Sensor

Steering Angle Sensor: The device operates within a voltage range of 8 V to 32 V DC and consumes 50 mA of current. It can function in temperatures between -30°C and $+85^{\circ}\text{C}$. Its measuring angle is $\pm 2160^{\circ}$, allowing for extensive rotational capability.



Figure 11.10: Steering Angle Sensor

11.6 Thermistors

Thermistor, is a temperature-sensitive resistor and they are used in sensing temperatures.

Negative temperature coefficient(NTC) Thermistor : In an NTC thermistor the resistance drops as temperature goes up. These are often used in sensing the temperature in cars.

Positive temperature coefficient(PTC) Thermistor : In a PTC thermistor the resistance increases with temperature. They are used in protection and control.

11.7 Tyre Pressure Monitoring System (TPMS)

Tyre Pressure Monitoring Systems is a device which keeps on monitoring air pressure inside vehicle tyres. Optimal tyre pressure can help improve mileage of vehicle and life of tyre as well as its safety. Generally passenger vehicles require a tyre pressure of about 35psi (pounds per square inch).

11.8 Radar and LiDAR Sensors

Modern vehicles employ the following sensors:

Radar(Radio Detection and Ranging) sensors work using microwaves in order to perform distance measurements and detect objects in close proximity.

LiDAR(Light Detection and Ranging) sensors use light (laser) to build up three dimensional information of the environment with high accuracy. Both types of sensors are widely utilized in automotive ADAS (advanced driver assistance systems).

CHAPTER 12

REGULATOR RECTIFIER (RR)

12.1 Introduction : The RR (Regulator Rectifier) is a vital automotive electronic unit used in two and three wheelers. The RR is essentially a dual functional unit. It performs the process of rectification where the AC voltage produced by the

alternator is converted to DC. The RR also serves the purpose of regulating the voltage at the output, ensuring it stays at a particular level necessary for the operation of the electrical loads and the charging of the battery.

12.2 Functions of Regulator Rectifier

Rectification: This function is used to convert the alternating current from the flywheel magneto into a direct current supply, this can then be used to charge the battery and power the vehicle's electric circuits.

Regulation: This is to regulate the voltage, so that it is constant, irrespective of the speed of the engine. The function of this is to prevent over-charging the battery and other electrical circuits.

12.3 Classification of Regulator Rectifiers

Depending on the type of supply to be regulated, Regulator Rectifiers can be classified as:

- Single Phase Regulator Rectifier
- Three Phase Regulator Rectifier

Single Phase Regulator Rectifier: Most two-wheelers utilize the single-phase RR systems. Some of the predominant ones are listed as:

- AC/DC Regulator Rectifier
- Full DC Regulator Rectifier
- Series Regulator Rectifier
- Shunt Regulator Rectifier

Three Phase Regulator Rectifier: Usually, the 3-phase RR systems are implemented in higher power applications. Some of the frequent ones are:

- Full DC Regulator Rectifier
- Series Regulator Rectifier
- Shunt Regulator Rectifier

12.4 Series Regulator Rectifier : In a Series Regulator Rectifier the control device is placed in series with the load and it allows only the current necessary for the load to pass to the load. For instance if the source is capable of providing 15A but the load demands only 5A the regulator allows 5A to pass to the load and nothing is drawn from the source. This yields to a lower power loss, higher efficiency.

12.5 Shunt Regulator Rectifier : In a Shunt Regulator Rectifier, the generator provides the full amount of available current. The load will draw only as much current as needed and the rest will be shunted to ground through the regulating device, generally a MOSFET. So if the generator supplies 15 A of current and the load needs 5 A, 10 A is shunted to ground. Shunt regulators are very easy to construct but do dissipate quite a lot of power.

12.6 Devices Used in RR

The power electronic devices that are incorporated in the Regulator Rectifiers are:

- Diodes
- Silicon Controlled Rectifiers (SCR)
- MOSFETs

The SCR is generally used for controlled rectifiers while MOSFET is preferred in the modern design of RR for high efficiency and rapid switching properties.



Figure 12.1: Regulator Rectifier

CHAPTER 13

MECHANICAL DESIGN CONCEPTS

13.1 Introduction

Mechanical design is an important part of automobile electronic product development. Designing a enclosure to protect and hold electronic circuits, including structural elements for mounting of electronic components and for providing resistance to the environment, reliability, strength, thermal dissipation. In INEL, I have been introduced to some fundamental principles associated with mechanical case design and manufacture during my internship.

13.2 Importance of Mounting

The mechanical system must be properly designed, and mounting is the crucial part which determines whether the PCB is positioned correctly and fastened securely to the mechanical casing. Correct mounting ensures mechanical system will be more resistant to vibration and more reliable and protects electronic components from mechanical stress during automotive application.

13.3 Mechanical Case Design Process : The PCB layout is translated to appropriate mechanical casing as per the mechanical size specification of the product. The following details are to be taken care of while designing the casing:

- Outline of the casing
- Identifying marks on the top and bottom side of the casing (silk print)
- Space requirement for track clearance
- Arrangement of supports at areas not populated by PCB tracks
- Mechanical constraints and assembly procedures

Normally the thickness of the casing is maintained close to 2 mm so as to obtain a rigid casing.

13.4 CNC Machining and Tool Design

Computer Numerical Control (CNC) machines, which is used to produce mold components with very high accuracy are utilized for the production of mold. The core and cavity design used in producing mechanical enclosures were considered for the production on mass scale.

13.5 Mass Production Processes

The most typical processes used in the manufacturing of the mechanical cases are;

- **Moulding**- used for acquiring the correct form of the enclosure
- **Anodizing**- used for increased resistance to corrosion and finishing
- **Shot Blasting**- used for increasing surface quality

All these processes help increase appearance and reliability of the product.

13.6 Rapid Prototyping

This technique called rapid prototyping is one method by which prototype models can be built and tested before mass production is started. It is used to check the design, fitment and functioning of the mechanical enclosure. Plastics are

normally preferred for this stage of development since they are easy to fabricate. The popular raw materials like filament and plastic pellets used in prototype manufacture.

13.7 Thermal Management in Mechanical Design

Thermal management is one critical design issue of automotive electronic products. The mechanical cases usually have cooling fins attached for heat dissipation from power semiconductor components such as MOSFET, SCR to the environment.

CHAPTER 14

INSTRUMENT CLUSTER AND DISPLAY TECHNOLOGIES

14.1 Introduction

In any modern vehicles, instrument cluster is the one which presents the essential information for the rider or driver about the operation of the vehicle. Instrument cluster takes the signals from various sensors and controllers and present the output like vehicle speed, engine rpm, fuel gauge, battery state, gear status and warn indicators etc. In INEL internship various type of instrument cluster and its associated communication methods was researched.



Figure 14.1: Istrument Cluster

14.2 Types of Instrument Clusters

Analog instrument cluster are the older forms of displaying vehicular information that uses a pointer or needle that points to the parameter displayed. The needles are driven by small motors which change the position as per the signals received from the sensors. The parameters are:

- Vehicle speed
- Engine RPM
- Fuel level
- Indicator lights (turn signals) etc. These are most commonly found in two wheelers.

Semi-Digital Instrument Cluster : A combination of analog and digital display technologies make up this kind of instrument cluster. Typically they have a combination of a needle gauge and a screen made out of LCDs. The analog portion of the cluster shows essential information like speed whereas the LCD portion provides:

- odometer readings
- gear selection displays
- fuel gauge levels
- engine speed (RPMs)

Digital LCD Instrument Cluster : All-digital clusters use a LCD screen (instead of analog needles) for all of the vehicle information. A microcontroller is receiving the sensor data and is also feeding information to the screen constantly. Common displayed information:

- Digital speedometer
- Digital tachometer

- Battery voltage
- Gear position
- Warning messages

TFT Instrument Cluster : TFTs generate high-resolution, colored graphics and are present in premium vehicle ranges. They are controlled by highly complex microcontrollers which are responsible for the display graphics and animations.

What's included:

- Navigation information
- Driving modes
- Text/call notifications
- Fuel level
- Warning signals

14.3 Communication Protocols Used in TFT Instrument Clusters

For TFT Instrument clusters high speed interfaces are necessary to quickly transfer the graphical data from the microcontroller to the display unit. These high speed interfaces are available in several variants depending on resolution, refresh rate and application requirements. Common communication protocols are:

1. RGB interface: Traditional display interface, where each color data red, green and blue is fed directly to the TFT display. Although simple, it is very sensitive to noise and requires high number of signals.

2. SPI (Serial Peripheral Interface): It is a synchronous serial interface widely used for low to medium resolution TFT displays. This interface needs only minimal connections.

3. IC (Inter-Integrated Circuit): this interface is widely used to configure the display controller and some auxiliary components as it supports low speed communication.

4. LVDS (Low Voltage Differential Signaling): It is a high speed differential communication interface mostly used in Automotive TFT clusters due to very good noise immunity, lower power consumption and to support high rate display data transfer.

5. MIPI (Mobile Industry Processor Interface): Latest high speed interface used for the recent display to support high resolution graphics and multimedia application. Of the above protocols, LVDS is the most widely used interface in Automotive TFT clusters.

14.4 CAN Communication in Instrument Clusters

Controller Area Network is a communication protocol which enables various electronic control units in a vehicle to transmit and receive information among them. Unlike previous signal transmissions where transistors were used, CAN uses less electrical noise and is more reliable. CAN messages received in the instrument cluster are about;

- Fuel level
- Speed of vehicle
- battery state
- gear state
- Warning

14.5 Display Inputs and Brightness Control The signals from the different vehicle inputs like head light on/off, fuel sensor readings, and other switches are fed into the microcontroller by utilizing the voltage divider circuits and transistor

based conditioning circuit. The brightness of the LEDs used in the instrument cluster is controlled by the PWM (Pulse Width Modulation) technique. The PWM technique can reduce the average power delivered to LEDs thus controlling the brightness of the LEDs.

CHAPTER 15

ENGINE FUEL INJECTION (EFI) SYSTEM AND EMISSION CONTROL

15.1 Introduction

Early vehicles relied on carburetors for supplying fuel. These carburetors are less precise as they fail to effectively measure fuel at varying engine speeds and consequently have greater emission levels and fuel economy. BS6 vehicles on the other hand use a fully electronically controlled fuel delivery system which is termed EFI, the Electronic Fuel Injection system, which controls the amount of fuel at the various engine working conditions.

15.2 Electronic Fuel Injection (EFI)

EFI system uses a system that electronically controls to inject the correct amount of fuel into the engine. This system makes better fuel efficiency, engine performance, and throttle response, also achieve the required emissions standards. To find out the injection quantity, the system uses the sensor inputs to analyse and decide it. Throttle body sensor:

The following sensor are normally present in the EFI throttle body:

1. Throttle position sensor.
2. Manifold Absolute Pressure sensor.

3. Intake Air Temperature sensor.

The sensor information is fed to the ECU and the ECU decide how much fuel has to be injected at different RPM.

15.3 Emission Control

The primary contents in exhaust gas of the car are :

- CO (Carbon Monoxide)
- HC (Hydrocarbons)
- CO₂ (Carbon Dioxide)
- NO_x (Nitrogen Oxides)

The current EFI system ensures the air-fuel ratio are being maintained an ideal mixture so as to obtain an optimal performance. The ideal air-fuel mixture ratio assures a good combustion to keep the emission rate down.

15.4 Air-Fuel Ratio

For petrol engines, the desired (stoichiometric) ratio of air to fuel is about 14.7:1, i.e., there are required 14.7 times as much air as fuel, and combustion is most efficient, with the least pollutants, with this ratio.

15.5 Dynamometer Testing

Dynamometer is an instrument used for testing vehicle performance and analyzing emission parameters under definite conditions to decide engine operating parameters and fuel injection to fulfill emission standards. Emission certification and testing procedures for automotive vehicles is done by organizations like ARAI, ICAT, GARC etc.



Figure 15.1: Dynamometer Testing

15.6 Transition from Carburetor to EFI

The carburetors in old vehicles were used to feed fuel to the engine by a mechanism. As carburetors can not effectively control the fuel supply according to the operating conditions so they were replaced by the EFI. Now in BS6 vehicles the carburetors are eliminated and are replaced by electronic throttle body and EFI system which improve the mileage, drivability and reduce emission.

CHAPTER 16

TESTING AND VALIDATION

To verify that an automotive electronic product is reliable, safe, and adheres to standards in the field, testing and validation have been implemented. This process determines whether the product meets the desired performance under various operating and environmental circumstances before it is ready for production. The product is verified according to specific requirements based on a Design Verification Plan (DVP). The significant tests being carried out are the following:

16.1 Performance Test: Performance testing is designed to test the functionality of the product under its operational condition. Resistive load is used as vehicle

load such as indicator lamp, warning lamp, and horn and so on, and its voltage, current and output response are observed to test the function of the product.

16.2 Endurance Test: Endurance test is to determine the long term reliability of the product by testing the product under continuously operating for a specific period. Failure is observed caused by wear out or material degradation and long-term operation.

16.3 Vibration Test: Electronic component of the automobile is constantly disturbed by vibration when the car is operating. Vibration test is designed to find out whether the component can endure mechanical stresses without having an impact on its performance and structure.

16.4 Salt Spray Test: Salt spray test is to determine corrosion resistance of metallic component or coating by exposure to controlled saline environment.

16.5 Electromagnetic Compatibility (EMC) Testing: Electronic product used in the automotive environment should work in the presence of electromagnetic disturbances. So EMC test should be performed. The following term has to be defined here:

EMI: Electromagnetic Interference. This is unwanted electromagnetic noise that emanates from an electronic device, interfering with and upsetting the operation of nearby devices.

EMS: Electromagnetic Susceptibility. This is the susceptibility of a device or equipment to suffer impaired performance when subjected to external electromagnetic disturbance.

EMC: Electromagnetic Compatibility. This is the ability of an electronic device or equipment to function satisfactorily in an electromagnetic environment without introducing electromagnetic interference to other devices or systems in that

environment, and without being disturbed by electromagnetic interference present in that environment. There are two types of EMC tests:

- **Conducted test** (disturbances transmitted through power and signal cables).
- **Radiated test** (disturbances transmitted through electromagnetic waves).

16.6 Testing Standards

Normally, below are the commonly applied automotive standards of electromagnetic and environmental test:

- ISO 11452 Road vehicles: Component test methods for electrical disturbances from narrowband radiated electromagnetic energy.
- ISO 61000 Electromagnetic compatibility requirements and test methods.
- AIS 004 Automotive Industry Standard related to EMC requirements for vehicles and components.
- CISPR 25 Limits and measurement methods for radio disturbance characteristics of vehicle components.

CHAPTER 17

CONCLUSION

The internship at India Nippon Electricals Limited (INEL), Hosur gave me an opportunity to learn the industrial aspect and the practical aspect of automotive electronic engineering. During the internship period, I got to know the design, manufacturing, testing and validation of the products used in automotive electronics. This internship allowed me to bridge the difference between my academic learning and industrial working environment.

I learned about different products that are used in automotive electronics like Flywheel Magneto, Ignition Coil, CDI, TCI, ECU, EV Controllers, Instrument Clusters, Regulator Rectifiers, EFI systems. I also learned about the manufacturing and designing of PCB using ORCAD for Schematic Development and using ECADSTAR and Allegro for PCB layout. I also learned about SMT and THT assembly techniques as well as about the protection methods of the components from ESD. The experience in Motor control systems, MOSFET-based power electronics and DC-DC converter gave me an exposure to the present automobile technology.

The analysis of Electronic circuit using LTspice helped me in improving my troubleshooting and analytical skills. The testing and validation of the product in automotive testing and validation for different tests like performance test, endurance test, vibration test, salt spray test, and EMC test helped me to gain an awareness of the reliability standards that is followed. Topics like EMI, EMS and EMC and their related standards like ISO 11452, ISO 61000, AIS 004 and CISPR 25 gave me an idea of the strict quality standards that are maintained by the automotive industry. I am confident that the skills acquired during this internship will serve as a firm basis for my future career and studies as a young Electronics and communication engineer.